Ioannis (John) Rizos

Professor of Theoretical Physics, Ph.D. — Ioannina, Greece

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Summary

Theoretical physicist with nearly thirty years of university teaching, examining and research supervision. Extensive practical experience in exactly the work that expert evaluation requires: authoring original physics and mathematics problems, writing and recording step-by-step model solutions, designing assessment items and marking rubrics, and judging whether a technical argument is correct and where it fails. Author of 71 peer-reviewed journal articles; referee for *Nuclear Physics B* and *JHEP* since 2008. Recent research applies machine learning, genetic algorithms and quantum annealing to large-scale classification problems.

Skills

Subject expertise. Quantum mechanics; quantum field theory; classical mechanics; electrodynamics; statistical physics; group theory; string theory and particle phenomenology. Mathematics for machine learning: linear algebra, calculus, probability, statistics, optimisation, numerical methods.

Evaluation and assessment. Original problem authoring; worked solution writing; multiple-choice item and distractor design; rubric construction; computer-based assessment systems; peer review of technical manuscripts; doctoral and master's thesis examination.

Technical. Python, Mathematica, FORTRAN, LaTeX; genetic algorithms, quantum annealing, machine-learning classification; UNIX/Linux.

Languages. Greek (native), English (fluent), French (very good).

Experience

Professor of Theoretical Physics — University of Ioannina 1996 – present

Assistant Professor 1996–2004; Associate Professor 2004–2014; Professor 2014–present

- Designed, taught and examined 15 undergraduate and graduate courses across quantum theory, classical mechanics, electrodynamics, group theory and computational physics — more than 120 semester-courses in total.
- Authored and marked written examinations and problem sets continuously since 1996, including the design of problems intended to discriminate between levels of understanding.
- Supervised 15 master's theses and one doctoral dissertation; served on numerous three-member advisory and seven-member doctoral examination committees.
- Created and taught *Computational Mathematics and Physics* for 22 semesters, covering numerical and symbolic computation in Python and Mathematica.

Problem Author and Scientific Editor — Papazisis Publications 3rd ed. 2019, 4th ed. 2022

Greek edition of Young & Freedman, Sears and Zemansky's University Physics with Modern Physics *(14th ed.)*

- One of a twelve-member academic team responsible for the scientific editing and adaptation of a two-volume, 1,880-page university physics textbook (3rd and 4th Greek editions).
- Wrote original physics problems not present in the American edition, and personally created and recorded model video solutions; the team produced 254 such solutions, delivered via QR codes printed in the text.

Tutor and Module Coordinator — Hellenic Open University 2004 – 2025

- *Mathematics for Machine Learning* (DAMA50), M.Sc. in Data Science and Machine Learning, taught in English, 2021–2025; Module Coordinator 2022–24. Linear algebra, calculus, probability, statistics and optimisation as the mathematical basis of machine learning.
- Member of the programme's Curriculum Committee, 2021–2024.
- *Quantum Physics* (2017–2021) and *Classical Physics II* (2004–2017); Module Coordinator for both. Distance education throughout, with written assessment as the primary mode.
- Authored multiple-choice question banks and produced 24 hours of recorded physics teaching material.

Technical Review, Evaluation and Examination 2008 – present

- Referee, on invitation, for *Nuclear Physics B*, *Journal of High Energy Physics*, *International Journal of Modern Physics A*, *Modern Physics Letters A* and *Universe* — assessing the correctness of technical arguments and derivations.
- Editorial boards: *International Journal of Modern Physics A* and *Modern Physics Letters A* (2020–), *Particles* (2021–).

- **Faculty appointment and promotion committees.** Member of three-person evaluation committees (*eisigitikes epitropes*) that assess candidates' full research records and produce the written, reasoned report on which appointment decisions are based — three such committees in 2025–26 alone, at the University of Ioannina, the Aristotle University of Thessaloniki and the NCSR Demokritos research centre. Also serves on the electoral bodies that decide academic appointments and promotions.

Leadership roles 2007 – present

- President, Southeastern European Network in Mathematical and Theoretical Physics (SEENET-MTP), 2018–present; re-elected 2025.
- Head, Department of Physics, University of Ioannina, 2013–2017; Deputy Head 2009–2013; Chair of the Internal Evaluation Group since 2022.

Assessment Systems and Education Research

- **TELEDIA —** co-developed a dynamic computer-based assessment system for university physics, in collaboration with the Department of Education (published 2002).
- Published studies on the evolution of physics students' conceptions of Newtonian mechanics, and on the use of digital technologies in assessment (2002–2007).

Education

Ph.D. in Physics — University of Ioannina, 1991. Thesis on the construction and phenomenological analysis of supersymmetric unified models from four-dimensional superstring theories.

B.Sc. in Physics — University of Ioannina, 1984.

Postdoctoral researcher, École Polytechnique, France (1991–93) and SISSA/ISAS, Italy (1993–96). Marie Curie Fellow (1995–96, 2003). Scientific Associate, CERN (2013).

Selected Publications

71 articles in peer-reviewed international journals; 5,886 citations, h-index 39 (Google Scholar). Most relevant to computational and machine-learning work:

- "String Model Building on Quantum Annealers," S. A. Abel, L. A. Nutricati and J. Rizos, *Fortschritte der Physik* 71 (2023) 2300167.
- "Towards Machine Learning in the Classification of $Z_2 \times Z_2$ Orbifold Compactifications," A. E. Faraggi, G. Harries, B. Percival and J. Rizos, *J. Phys.: Conf. Ser.* 1586 (2020) 012032.
- "Genetic Algorithms and the Search for Viable String Vacua," S. Abel and J. Rizos, *JHEP* 1408 (2014) 010.
- "MERLIN: A Portable System for Multidimensional Minimization," G. A. Evagelakis, J. Rizos, I. E. Lagaris and I. N. Demetropoulos, *Computer Physics Communications* 46 (1987) 401–415.

Full list: inspirehep.net/authors/991609